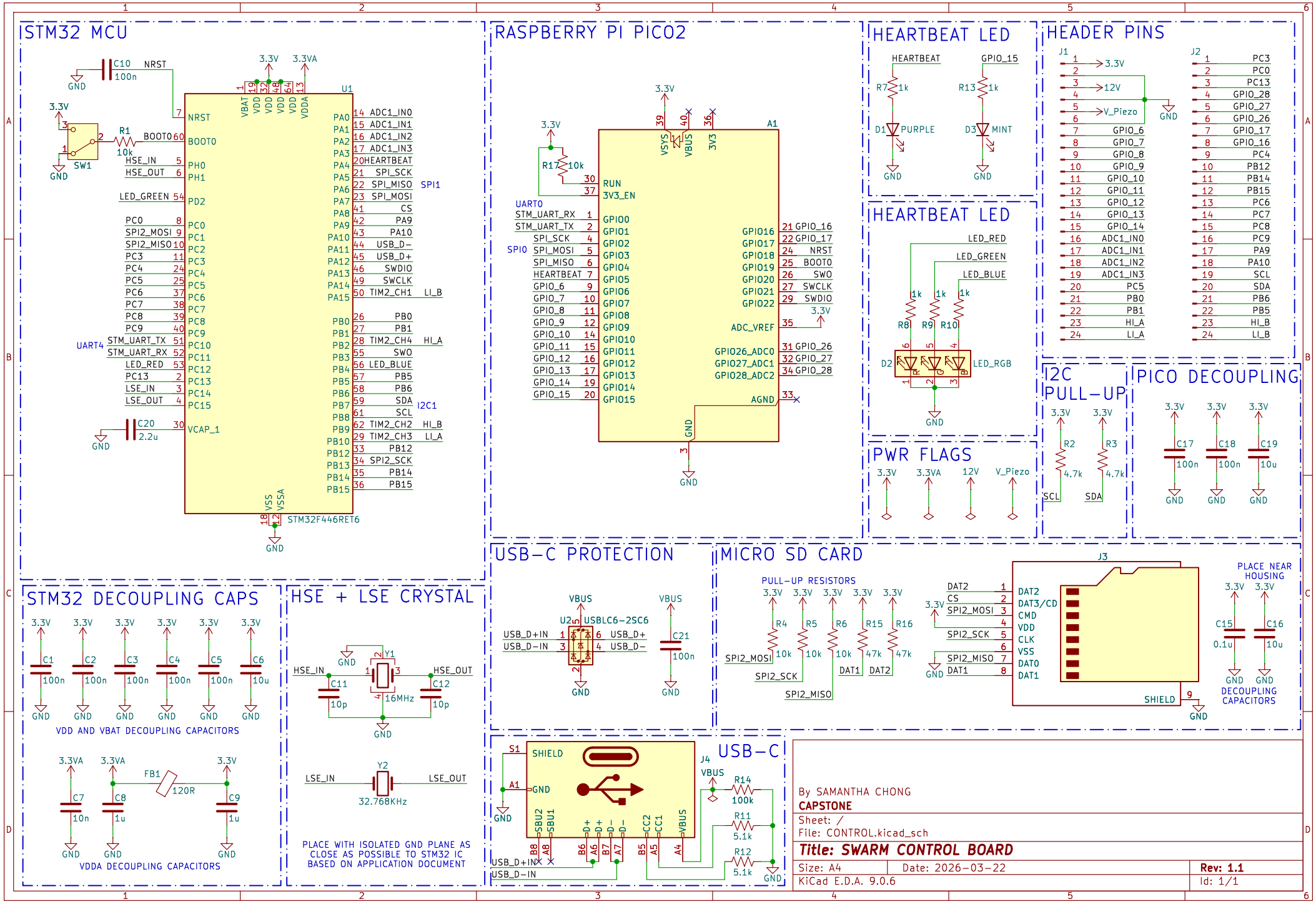
STM32 MCU
U1
STM32F446RET6
NRST
BOOT0
HSE_IN
HSE_OUT
LED_GREEN
UART4
STM_UART_TX
STM_UART_RX
LED_RED
LED_BLUE
SPI1
SPI2_MOSI
SPI2_MISO
SPI2_SCK
I2C1
SCL
SDA
ADC1_IN0
ADC1_IN1
ADC1_IN2
ADC1_IN3
HEARTBEAT
USB_D-
USB_D+
SWDIO
SWCLK
TIM2_CH1
TIM2_CH4
TIM2_CH2
TIM2_CH3
LI_B
HI_A
HI_B
LI_A
RASPBERRY PI PICO2
A1
RUN
3V3_EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
VSYS
VBUS
3V3
HEARTBEAT LED
HEARTBEAT
GPIO_15
PURPLE
MINT
HEADER PINS
J1
J2
PWR FLAGS
3.3V
3.3VA
12V
V_Piezo
I2C PULL-UP
R2
R3
4.7k
PICO DECOUPLING
C17
C18
C19
100n
10u
STM32 DECOUPLING CAPS
VDD AND VBAT DECOUPLING CAPACITORS
VDDA DECOUPLING CAPACITORS
FB1
120R
HSE + LSE CRYSTAL
Y1
16MHz
Y2
32.768KHz
PLACE WITH ISOLATED GND PLANE AS CLOSE AS POSSIBLE TO STM32 IC BASED ON APPLICATION DOCUMENT
USB-C PROTECTION
U2
USBLC6-2SC6
USB_D+IN
USB_D-IN
USB-C
J4
S1
SHIELD
MICRO SD CARD
J3
PULL-UP RESISTORS
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
PLACE NEAR HOUSING
DECOUPLING CAPACITORS
By SAMANTHA CHONG
CAPSTONE
Sheet: /
File: CONTROL.kicad_sch
Title: SWARM CONTROL BOARD
Size: A4
Date: 2026-03-22
Rev: 1.1
KiCad E.D.A. 9.0.6
Id: 1/1